

Stock Market Monitoring and Alert System using n8n

Problem Statement

Investors and traders need to continuously monitor stock prices and market fluctuations to make informed investment decisions. Manually checking stock prices from financial websites throughout the day is time-consuming and inefficient. There is a need for an automated system that can periodically collect stock market information and instantly notify users about the latest stock updates.

Proposed Solution

The proposed system uses n8n to automate the process of monitoring stock market data. The workflow fetches real-time stock information from the Alpha Vantage API at scheduled intervals and sends the latest stock details directly to users through Telegram. This automation reduces manual effort and ensures timely delivery of market information.

Objectives

- Automate stock market monitoring.
- Retrieve real-time stock information using APIs.
- Send automated Telegram notifications.
- Reduce manual effort in checking stock prices.
- Demonstrate workflow automation using n8n.

Workflow

Schedule Trigger → HTTP Request (Alpha Vantage API) → Telegram Notification

Tools and Technologies Used

- n8n
- Alpha Vantage API

- Telegram Bot API
- Docker
- HTTP Request Node
- Schedule Trigger Node

Expected Outcome

The system automatically fetches stock market data at regular intervals and sends the latest stock information to the user through Telegram. This enables users to stay updated about stock market movements without manually visiting financial websites.

Advantages

- Fully automated workflow.
- Real-time stock monitoring.
- Instant Telegram notifications.
- Easy to configure and maintain.
- Saves time and effort.

Future Enhancements

- Monitor multiple stocks simultaneously.
- Add Indian stock market support.
- Implement stock price prediction using Machine Learning.
- Create a dashboard for visualizing stock trends.
- Add email and WhatsApp notifications.